# Toward a four-dimensional Yang–Mills construction

Notebook 35: infrared cancellation and physical heat bounds

Research working document

17 October 2026

**Abstract**

We derive an exact local, gauge-covariant representation of the actual leading determinant using zero covariant block means in the scalar gauge. Both its scalar and vector factors have flat spectral gaps of order $K^{-2}$, with uniform heat and inverse-power bounds. The finite amplitude identity continues jointly through the flat stabilizers and passes to local infinite systems at each fixed $K$. Its invariant diagonal densities give the actual Maxwell moment directly. This closes the specific infrared passage left open by the earlier massless physical quotient.

The exact finite physical quotient of Notebook 34 requires joint control of its scalar and physical determinants. We give an explicit reason: a globally based massless scalar determinant on a periodic constant Cartan connection has a nonzero quadratic response per volume, whereas the ordered thermodynamic and zero-mass limit gives zero. The fine/coarse scalar ratio has an extra term proportional to $K^2 - 1$. The actual local gauge Ward identity and the finite quotient formula prove that the physical determinant cancels this term exactly in the thermodynamic limit. No flux seam or nonperiodic magnetic family is used.

We also extend the flat physical spectral estimates to uniform heat-trace and inverse-power bounds. The rescaled inverse square has at most logarithmic trace growth; powers above two have uniformly bounded trace per final block. A local tent lift proves full Laplacian domination for the new vector gauge, so both new factors have the corresponding estimates. Their endpoint-uniform curvature comparison remains the next step toward a positive actual drift.

## Scope and conventions

All actual operators are those of the homogeneous local construction in Notebooks 32–34. The primitive blocking ratio is fixed, $K = \ell^k$, and $M$ denotes the preceding homogeneous depth. Traces count real adjoint coordinates. For a normalized Cartan element $H$, $\tau(H^2) = 1$, set

$$C_H = \sum_{q \in \mathrm{spec}(\mathrm{ad}_H)} q^2, \qquad \alpha_g = \frac{C_H}{48\pi^2}.$$

The complexified charge decomposition counts each real adjoint degree of freedom once in this trace convention. The notation $[t^2]$ denotes a Taylor coefficient, not a second derivative.

The constant flat connection used below tests the zero-momentum Ward identity. It is different from the constant-curvature probe that extracts the Maxwell coefficient. The infrared order of limits is stated explicitly for each result.

## 1 The scalar infrared limit and a forced holonomy cancellation

There are two different assertions in this section. A common positive physical mass gives a well-defined thermodynamic scalar reference, whose subsequent massless limit is $\alpha_g \log K$. The original separately

massless globally based determinants need not have that order-independent analytic limit. An exactly periodic flat connection gives a concrete obstruction and, through the actual finite determinant identity, forces a compensating response in the physical factor.

### 1.1 The mass convention and an exact base correction

Fix $K = \ell^k$ and final torus side $N$ in four dimensions. Current fine sites have physical spacing $K^{-1}$, with side $KN$; coarse sites have spacing one and side $N$. All color coordinates are trace orthonormal. Let $G_{\mathrm{all}}(U)$ be the covariant scalar Laplacian on all sites, and let $G_b(U)$ be its principal minor after deleting the color coordinates at one common physical base $o$. Deleting a site does not change the remaining diagonal entries.

A common physical mass $\delta > 0$ means

$$H_{f,\delta} = K^2 G_{\mathrm{all},f} + \delta^2 I, \qquad H_{c,\delta} = G_{\mathrm{all},c} + \delta^2 I. \tag{1}$$

Their principal minors are $H^b_{f,\delta}, H^b_{c,\delta}$. With $\mathcal{N}_0 F = F(W) - F(1)$, define

$$S^b_{\delta,N} = \frac{1}{2}\mathcal{N}_0(\log\det H^b_{f,\delta} - \log\det H^b_{c,\delta}). \tag{2}$$

The fine counting mass is $\delta/K$, not $\delta$. At zero mass multiplication of the fine operator by $K^2$ changes only a background-independent normalization. Equation (2) is a specified scalar comparison family. It is not automatically an exact mass deformation of the entire physical-quotient determinant identity.

**Lemma 1.1** (Positive-mass base defect)**.** *Put $H = \epsilon^{-2} G_{\mathrm{all}} + \delta^2 I$, $R = H^{-1}$, where $\epsilon = K^{-1}$ or $1$. Then*

$$\det H^b = \det H \, \det R_{oo}. \tag{3}$$

*At fixed $\delta > 0, K$, the normalized logarithm of the fixed color block $R_{oo}$ has a complex product neighborhood and first-two bounded-direction derivative bounds independent of $N$. Its first-two source kernels are exponentially summable in their distances from $o$, including all fixed polynomial moments. No uniformity as $\delta \downarrow 0$ is asserted.*

*Proof.* The determinant identity is the block Jacobi identity. At real backgrounds,

$$\delta^2 I \le H \le (16\epsilon^{-2} + \delta^2) I, \qquad (16\epsilon^{-2} + \delta^2)^{-1} I \le R_{oo} \le \delta^{-2} I.$$

The nearest-neighbor coefficients have volume-independent operator bounds on a small complex product ball. The resolvent identity, and then the lower bound for $R_{oo}$, give a smaller complex ball on which both inverses remain bounded uniformly in volume. The logarithm branch continued from zero is bounded there after normalization. Its first two derivatives also follow directly from

$$\begin{aligned} DR[v] &= -RH'[v]R, \\ D^2R[v,w] &= RH'[v]RH'[w]R + RH'[w]RH'[v]R - RH''[v,w]R. \end{aligned} \tag{4}$$

Differentiating $\log\det R_{oo}$ uses only its fixed color trace.

For exponential bounds, conjugate by a site weight whose ratios across nearest-neighbor edges differ from one by $O(\kappa)$. Choose $\kappa > 0$, depending on $\delta, \epsilon$, so the conjugation perturbation has norm at most $\delta^2/4$. The weighted inverse is then bounded by $2/\delta^2$. Spending a smaller exponent gives absolute row and column Schur bounds. The derivatives $H', H''$ are local. Every source chain in (4), and in the logarithmic derivatives, begins and ends at $o$. Its exponential weights therefore control all source distances from $o$, with a further exponent reserve for any fixed polynomial moment. □

Writing $S^{\mathrm{all}}_{\delta,N}$ for the unbased half-ratio gives the exact comparison

$$S^{b}_{\delta,N} = S^{\mathrm{all}}_{\delta,N} + \frac{1}{2}\mathcal{N}_0\{\log\det(R_f)_{oo} - \log\det(R_c)_{oo}\}. \tag{5}$$

The lemma has two useful consequences. A base receding from fixed localized sources changes their first-two coefficients exponentially little. Alternatively, average the based functional over the $N^4$ simultaneous coarse translations of the fine and coarse base. Reanchoring its Hessian at one output cell bounds the anchored second-moment norm of its difference from the unbased functional by

$$C_{\delta,K}N^{-4}. \tag{6}$$

Indeed summing the translated rooted defect kernel converts that row into its full source sum, divided by $N^4$; its displacement moments are bounded by the lemma. This averaging is an estimate for the limit, not a claim that different finite base choices give identical functionals.

After composition with the actual fixed-$K$ stationary branch, the same conclusions use its previously proved localized source kernels. Under the Notebook 33 inputs the constants are uniform in old depth $M$, still depending on $K, \delta$.

### 1.2 The ordered localized Maxwell limit

At positive mass, the unbased diagonal logarithm has exponentially localized Taylor kernels, by the same resolvent argument applied to its logarithm integral. Its color trace is gauge invariant. Its Hessian moments therefore define its affine curvature coefficient absolutely. This gives a way to take the limit without postulating an affine magnetic field on a torus: first pass to infinite kernels using compactly supported directions, then evaluate polynomial directions by their exponential moments. Symmetric short lifts of periodic kernels give the same result. At fixed $K, \delta$, unbased winding contributions vanish exponentially in $N$, and (6) removes the averaged base contribution.

For an actual stationary insertion, use fixed-$K$ structural compactness followed by the polynomial-jet identification of Notebook 33. Every such limit has the same scalar expression below. This statement uses the already established source kernels; it does not infer locality from a finite determinant identity.

Normalize the Maxwell functional by $\beta(F) = 2[t^2]$ of its localized constant-curvature density. For $\mu \neq \nu$ and $\tau(H^2) = 1$, the current and coarse Cartan probes are

$$U_\mu(t;x) = e^{itx_\nu H/K^2}, \qquad W_\mu(t;y) = e^{ity_\nu H}.$$

The unpinned scalar heat coefficient already established is

$$\begin{aligned} h(s) &= \frac{C_H}{3}s^2e^{-8s}I_0(2s)^2I_1(2s)^2, \\ h(s) &= O(s^4) \quad (s \downarrow 0), \\ h(s) &= \alpha_g + O(s^{-1}) \quad (s \to \infty), \qquad \alpha_g = \frac{C_H}{48\pi^2}. \end{aligned} \tag{7}$$

The $K^4$ fine sites per final block cancel the curvature factor $K^{-4}$. In common physical heat time $u$, the fine and coarse coefficients are consequently $h(K^2u)$ and $h(u)$.

**Proposition 1.2** (Ordered scalar reference limit)**.** *Take the thermodynamic localized-kernel limit first, at fixed $\delta > 0$ and fixed $K$. The resulting scalar coefficient is*

$$\beta_\delta(K) = \frac{1}{2}\int_0^\infty e^{-\delta^2 u}\{h(K^2u) - h(u)\}\frac{du}{u}. \tag{8}$$

*It satisfies*

$$\lim_{\delta\downarrow 0}\beta_\delta(K)=\alpha_g\log K. \tag{9}$$

*For $K\geq 1$ and $0<\delta\leq 1/2$, the explicit reference also obeys*

$$|\beta_\delta(K)-\alpha_g\log K|\leq C\delta^2\{1+\log(1/\delta)\}, \tag{10}$$

*where $C$ is independent of $K$.*

*Proof.* The localized logarithm heat formula, the preceding normalization, and (7) give (8). Without the mass factor, the paired integrand is absolutely integrable: use its small-$s$ bound at zero and cancellation of $\alpha_g$ at infinity. On $\varepsilon<u<T$, its unhalved integral equals

$$\int_T^{K^2T}h(v)\frac{dv}{v}-\int_\varepsilon^{K^2\varepsilon}h(v)\frac{dv}{v}.$$

The limits are $2\alpha_g\log K$ and zero, respectively. Dominated convergence proves (9).

For the rate, $h$ is bounded on the whole positive axis. On $0<u<1$, use $1-e^{-\delta^2u}\leq\delta^2u$; the resulting error is $O(\delta^2)$, uniformly in $K$. For $u\geq 1$, (7) gives $|h(K^2u)-h(u)|\leq 2C/u$ for every $K\geq 1$. Split the remaining integral at $u=\delta^{-2}$. The first part is $O(\delta^2\log(1/\delta))$, the second $O(\delta^2)$. □

The uniformity in (10) belongs to the explicit reference after the thermodynamic limit. It does not interchange mass removal with that limit for the original based determinants.

### 1.3 An exact massless anomaly on a periodic flat connection

There is a direct obstruction to such an interchange, already at zero momentum. Consider one charged scalar $q$ on a periodic $n^4$ lattice, with constant phase $e^{itqA}$ on direction $\mu$. This is a legitimate periodic flat connection for every $t$; no continuously varying magnetic flux is involved. Its unbased eigenvalues are

$$\lambda_p(t)=2\sum_\nu\{1-\cos(p_\nu+tqA\,\delta_{\mu\nu})\},\qquad p\in(2\pi/n)\mathbb{Z}_n^4. \tag{11}$$

Put $V=n^4$, and delete one scalar site to obtain $G_n^b(t)$. Translation invariance of the resolvent diagonal and (3) imply

$$\det G_n^b(t)=\frac{1}{V}\prod_{p\neq 0}\lambda_p(t)\left\{1+\lambda_0(t)\sum_{p\neq 0}\lambda_p(t)^{-1}\right\}. \tag{12}$$

Initially this is an identity away from zero eigenvalues; both sides continue analytically through $t=0$.

**Proposition 1.3** (Noncommuting mass and thermodynamic derivatives)**.** *Let*

$$g_4=\int_{[-\pi,\pi]^4}\frac{dp}{(2\pi)^4}\frac{1}{2\sum_\nu(1-\cos p_\nu)}>0. \tag{13}$$

*Then*

$$\lim_{n\to\infty}n^{-4}[t^2]\log\det G_n^b(t)=q^2A^2g_4. \tag{14}$$

*At every fixed positive scalar mass, the corresponding thermodynamic coefficient is zero, and remains zero when the mass is subsequently removed.*

*Proof.* Since $\lambda_0(t) = q^2A^2t^2 + O(t^4)$, (12) gives, writing $\lambda_p = \lambda_p(0)$,

$$[t^2] \log\det G_n^b(t) = q^2A^2 \sum_{p\neq 0} \left\{ \frac{\cos p_\mu}{\lambda_p} - \frac{2\sin^2 p_\mu}{\lambda_p^2} + \frac{1}{\lambda_p} \right\}. \tag{15}$$

The first two summands equal $\partial_{p_\mu}(\sin p_\mu/\lambda_p)$. Their singularity is $O(|p|^{-2})$, integrable in dimension four. Removing a radius-$\varepsilon$ ball before integration by parts leaves boundary flux $O(\varepsilon^2)$, so their Brillouin-zone integral is zero. Their grid sums converge:

$$n^{-4} \sum_{0<|p|\leq\varepsilon} |p|^{-2} \leq C\varepsilon^2 + O(n^{-2}). \tag{16}$$

Outside that ball ordinary Riemann convergence applies. The same argument for $\lambda_p^{-1}$ gives $g_4$, proving (14).

At fixed positive mass $m$, the unbased coefficient per volume instead tends to the integral of $\partial_{p_\mu}\{\sin p_\mu/(\lambda_p + m^2)\}$, which is zero by periodicity. Lemma 1.1 makes the based correction $O(1)$ in these derivatives, so division by $n^4$ removes it. Subsequent $m \downarrow 0$ still gives zero. □

The deleted block has fixed rank, but its massless derivative contribution is extensive. Equation (12) identifies the soft global mode responsible for the failure. Summing adjoint charges replaces $q^2$ by $C_H$; for a vector of commuting constant angles the answer is $C_H|A|^2 g_4$, with mixed terms zero by coordinate reflections.

### 1.4 The actual branch and its forced physical compensation

Now use the actual final constant links and current representative

$$W_\mu(t) = e^{itA_\mu H}, \qquad U_\mu(t) = e^{itA_\mu H/K}. \tag{17}$$

Each ordinary blocking stage preserves this commuting connection: all paths with the same displacement have the same transporter, so the selected mean is evaluated on identical entries. The same flat lift exists through every old refinement $M$. All plaquettes are identity. Since the real Wilson action is nonnegative, the associated classical minimum values are zero. After the actual local root gauge, uniqueness of the small minimum germ identifies this with the actual stationary branch. This is a fixed finite-germ argument.

Every sitewise adjoint gauge preserves a scalar zero at the global base. Thus the globally based scalar determinants are unchanged by that root gauge and may be computed in (17). Fine current volume is $(KN)^4$ and fine angle is $tA/K$; coarse volume is $N^4$ and angle is $tA$. Proposition 1.3 therefore gives

$$\lim_{N\to\infty} N^{-4}[t^2](\log\det G_f - \log\det G_c) = C_H|A|^2(K^2-1)g_4. \tag{18}$$

The generators on this branch are the same for every old $M$. This is a flat-holonomy, zero-momentum response, not a Maxwell coefficient.

**Proposition 1.4** (Forced compensation in the physical factor)**.** *Fix $K$ before taking the thermodynamic limit. For the actual leading correction and physical quotient,*

$$\lim_{N\to\infty} N^{-4}[t^2] D_N(W(t)) = 0, \tag{19}$$

$$\lim_{N\to\infty} N^{-4}[t^2] \log\det Q_\perp(W(t)) = C_H|A|^2(K^2-1)g_4. \tag{20}$$

*The same conclusion allows varying old depth under the previously established uniform-old-depth, fixed-$K$ local kernel bounds.*

*Proof.* The actual correction $D_N$ has translation-covariant Hessian kernels with common exponential row and source bounds at fixed $K$. Every structural limit has invariant local densities and zero linear color jet. Its quadratic Ward identity consequently annihilates lattice gradients. Exponential moments extend this identity to linear scalar gauge parameters by cutoff approximation. Their gradients are constant link fields, so the infinite quadratic symbol vanishes at momentum zero.

The finite normalized coefficient on (17) is one-half the contracted Hessian row sum. Uniform exponential tails make its distant part uniformly small. Every fixed central sum converges along a structural subsequence, and its full limiting row sum is zero by the preceding Ward argument. All subsequential scalar limits are therefore zero, proving (19). The uniform-old-$M$ versions of these same bounds permit $M$ to vary. No uniform-in-$K$ locality is used.

At finite $N$ the exact physical-quotient identity is

$$2D_N = \mathcal{N}_0(\log\det G_f - \log\det G_c - \log\det Q_\perp). \tag{21}$$

Subtract (19) from (18) using the same branch, base and regulator. This yields (20), including its sign and normalization. □

No claim that $D_N(W(t))$ is identically zero at finite $N$ is needed: a periodic constant connection may have nontrivial holonomy. Nevertheless, its thermodynamic constant-field response must vanish by locality and the Ward identity. Thus the term $-\frac{1}{2}\log\det Q_\perp$ cancels the scalar half-ratio exactly at zero momentum.

### 1.5 What the ordered limit does not identify

The positive result is

$$\lim_{\delta\downarrow 0}\lim_{N\to\infty}\{\text{localized regulated scalar Maxwell coefficient}\} = \alpha_g \log K.$$

The original separately massless globally based family does not have an order-independent analytic local limit, as the periodic test (18) proves. In particular a $K$-uniform $O(1)$ constant-potential response for the separate physical determinant is false, despite its flat lower bound $cK^{-2}$.

The anomaly alone neither computes nor disproves equality of Maxwell coefficients after a suitable coupled cancellation. A valid joint vector/scalar regulator must preserve the rank-fixed but derivative-large base factors before its infrared limit. The exact finite quotient identity does not provide that regulated identity by itself. No continuum measure, nontriviality, or mass-gap conclusion follows here.

## 2 Flat physical heat densities and the remaining curvature interface

Fix dimension four, a color algebra of dimension $n_{\mathfrak{g}}$, a primitive odd ratio $\ell$, a composite $K = \ell^k$, and any preceding homogeneous depth $M$. Coordinates are current-link angles with counting inner products. On a cubic current torus of side $KN$, there are $N^4$ final $K$-blocks. All statements about varying backgrounds below are explicitly separated from the flat spectral theorem.

### 2.1 Actual flat inputs

Let $P_f, P_c$ be the fine and coarse physical projections at the flat field, and write

$$\begin{aligned} &L = q_M, \qquad A = P_c C_K|_{\operatorname{ran} P_f}, \\ &R = L|_{\operatorname{ran} P_f} - A^T q_{MK} A, \qquad Q = R + aA^T A. \end{aligned} \tag{22}$$

These are the actual residual physical operators of the finite quotient identity. The following previously proved facts are the only model-specific inputs:

$$\begin{aligned} L \ge d^T d, \qquad & 0 \le q_{MK}, \qquad \|q_{MK}\| \le R_q, \\ R \ge 0, \qquad & Q \ge c_a K^{-2} I, \qquad c_a > 0. \end{aligned} \tag{23}$$

The constants are uniform in $M, K, N$. For example, the physical alias proof gives

$$c_a = \min\{B_0^{-1}, a/(A_0 + B_0 R_q)\}, \qquad A_0 = 2c_\omega^{-2}, \quad B_0 = (1 + 2c_\omega^{-2})/4, \quad c_\omega = (2/\pi)^5.$$

Flat operators are invariant under final-block translations.

The globally based scalar gradient has the same range as the ordinary periodic gradient: subtracting the value of a scalar at the base does not change its gradient. Thus $\operatorname{ran} P_f$ is the usual transverse Bloch space, including all harmonic one-forms at momentum zero. No estimate on an inverse globally based scalar operator is used.

Choose a fixed $a \ge R_q$. Then

$$Q = L|_{\mathrm{phys}} + A^T(aI - q_{MK})A \ge \Delta_{\mathrm{phys}}, \qquad Q \ge \tfrac{1}{2}\Delta_{\mathrm{phys}} + \frac{c_a}{2K^2} I. \tag{24}$$

Here $\Delta_{\mathrm{phys}}$ is the Maxwell form on the transverse space, where it is the scalar lattice Laplacian.

**Theorem 2.1** (Uniform flat physical heat and inverse moments)**.** *Put $\widehat{Q} = K^2 Q$, and define the heat trace per final block by*

$$\widehat{H}_{M,K,N}(u) = N^{-4} \operatorname{Tr}_{\mathrm{phys}} e^{-u\widehat{Q}}.$$

*There are $C, c > 0$, independent of $M, K, N$, such that*

$$\widehat{H}_{M,K,N}(u) \le C \left\{ \frac{K^4}{(1 + K^2 u)^2} + N^{-4} \right\} e^{-cu}, \qquad u > 0. \tag{25}$$

*In the infinite flat structural system the corresponding block trace satisfies the same bound with $N^{-4}$ omitted. For every $p > 0$,*

$$\operatorname{Tr}_{\mathrm{block}} \widehat{Q}^{-p} \le C_p \begin{cases} K^{4-2p} + N^{-4}, & 0 < p < 2, \\ 1 + \log K + N^{-4}, & p = 2, \\ 1 + N^{-4}, & p > 2, \end{cases} \tag{26}$$

*again omitting $N^{-4}$ in infinite volume. For each fixed real $r$ and $U \ge 1$,*

$$\int_U^\infty u^{r-1} \widehat{H}_{M,K,N}(u)\, du \le C_r (1 + N^{-4}) e^{-cU/2}. \tag{27}$$

*These are unperturbed spectral statements. They do not assert a bound on a background derivative of a heat trace.*

*Proof.* For self-adjoint finite matrices $X \ge Y$, the min–max theorem gives ordered eigenvalues $\lambda_j(X) \ge \lambda_j(Y)$; therefore $\operatorname{Tr} e^{-sX} \le \operatorname{Tr} e^{-sY}$. No operator monotonicity of the exponential is asserted.

At momentum $p \in (2\pi/n)\mathbb{Z}_n^4$, $n = KN$, put $P(p) = 4\sum_\mu \sin^2(p_\mu/2)$. The physical Maxwell multiplicity is $3n_{\mathfrak{g}}$ away from zero and $4n_{\mathfrak{g}}$ at zero. Hence

$$\operatorname{Tr}_{\mathrm{phys}} e^{-sQ} \le 4n_{\mathfrak{g}} e^{-c_a s/(2K^2)} \sum_p e^{-sP(p)/2}. \tag{28}$$

For the one-dimensional scalar heat average

$$p_n(t) = \frac{1}{n} \sum_{m \bmod n} e^{-4t \sin^2(\pi m/n)},$$

use representatives $|m| \le n/2$ and $\sin(\pi|m|/n) \ge 2|m|/n$. Gaussian summation for $t \ge 1$, and $p_n(t) \le 1$ for $t \le 1$, give

$$p_n(t) \le n^{-1} + C(1+t)^{-1/2}, \qquad p_n(t)^4 \le C\{n^{-4} + (1+t)^{-2}\}. \tag{29}$$

Since $\sum_p e^{-tP(p)} = n^4 p_n(t)^4$, division of (28) by $N^4$ yields

$$N^{-4} \operatorname{Tr} e^{-sQ} \le C\{K^4(1+s)^{-2} + N^{-4}\} e^{-cs/K^2}.$$

Set $s = K^2 u$ to obtain (25).

The normalized total trace equals the diagonal trace in any one final block, by final-block translation invariance. This equality does not turn the min–max argument into a pointwise operator comparison. In infinite volume, apply the same min–max argument in each finite-dimensional coarse Bloch fiber and integrate. Folding the Maxwell aliases gives $K^4$ times the fine scalar heat density, proving the infinite statement directly.

Positive spectral calculus gives

$$\operatorname{Tr}_{\text{block}} \widehat{Q}^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty u^{p-1} \operatorname{Tr}_{\text{block}} e^{-u\widehat{Q}}\, du.$$

Split at $K^{-2}$ and 1 to obtain (26). For $p > 2$, one can instead use $K^4/(1+K^2u)^2 \le u^{-2}$ directly. The same inequality and exponential decay prove (27). □

The finite $N^{-4}$ term is necessary: it retains the finite number of physical harmonic modes. The local inverse moment means $\operatorname{Tr}(P_{\text{block}} \widehat{Q}^{-p} P_{\text{block}})$. It is not a claim of global trace class in infinite volume, nor an identification with a power of the compressed inverse. The logarithm in the $p = 2$ bound is an upper bound; no matching actual asymptotic is claimed.

### 2.2 An optional finite on-shell replacement

On a finite real on-shell family with $LD_f = 0$ and $qD_c = 0$, set

$$Q_{\text{on}} = L|_{\text{phys}} + aA^T A = Q + A^T q_{\text{phys}} A.$$

The minimum right inverse gives $AQ^{-1}A^T = a^{-1}I$. Consequently the determinant lemma proves

$$\frac{\det Q_{\text{on}}}{\det Q} = \det(I + a^{-1} q_{\text{phys}}). \tag{30}$$

Because $q$ kills the coarse gauge space, the normalized logarithm on the right is the normalized full coarse $\log\det(q + aI)$; only a field-independent power of $a$ has been removed. The already proved coarse-factor Maxwell estimate can therefore control this replacement wherever its Maxwell moment is legitimately defined.

This does not apply to an arbitrary off-shell family by omission of the multiplier curvature. The actual general form is $L = s'' - \lambda \cdot f''$. An infinite affine use of separate globally based physical determinants requires its own infrared justification.

### 2.3 A precise sufficient derivative estimate

The following fixed-space lemma isolates an additional estimate which would suffice for an infrared second-variation bound. Its hypotheses are not established here for the actual varying physical quotient.

**Lemma 2.2** (Relative-form vertices)**.** *Let $X(t) = X + tV_1 + t^2V_2 + O(t^3)$ be a real self-adjoint family on a fixed finite-dimensional Hilbert space, with $X > 0$. Suppose*

$$\|X^{-1/2}V_jX^{-1/2}\| \le L_j, \qquad j = 1, 2. \tag{31}$$

*Then*

$$\left|[t^2] \operatorname{Tr} e^{-uX(t)}\right| \le 8(L_2 + L_1^2) \operatorname{Tr} e^{-uX/4}, \qquad u > 0. \tag{32}$$

*Proof.* The single insertion has absolute value at most

$$uL_2 \operatorname{Tr}(Xe^{-uX}) \le (2/e)L_2 \operatorname{Tr} e^{-uX/2}.$$

In an eigenbasis of $X$, cyclic Duhamel gives the double insertion exactly as

$$\frac{u}{2} \sum_{i,j} |(V_1)_{ij}|^2 \frac{e^{-u\lambda_i} - e^{-u\lambda_j}}{\lambda_j - \lambda_i}, \tag{33}$$

with the continuous value $ue^{-u\lambda_i}$ at coincident eigenvalues. For $a, b > 0$,

$$ab\,\frac{e^{-ua} - e^{-ub}}{b - a} \le 2\{ae^{-ua/2} + be^{-ub/2}\}. \tag{34}$$

Indeed, assume $b \ge a$. For $b \ge 2a$, use $ab/(b - a) \le 2a$. For $a \le b \le 2a$, use the derivative bound $ue^{-ua}$, $ab \le 2a^2$, and $(ua)e^{-ua/2} \le 2/e$.

Write $V_1 = X^{1/2}BX^{1/2}$, with $\|B\| \le L_1$. The row and column square sums of $B$ are at most $L_1^2$. Applying (34) to (33) bounds it by

$$2uL_1^2 \operatorname{Tr}(Xe^{-uX/2}) \le (8/e)L_1^2 \operatorname{Tr} e^{-uX/4}.$$

Together with the single insertion this proves the stated conservative constant. □

If $X(t) = K^2Q_\perp(t)$ could be identified with a fixed physical space and satisfy (31) uniformly, Theorem 2.1 would give

$$\int_1^\infty \left|N^{-4}[t^2] \operatorname{Tr} e^{-uX(t)}\right| \frac{du}{u} \le C(L_2 + L_1^2)(1 + N^{-4}). \tag{35}$$

An infinite-volume version additionally requires a justified trace-and-derivative limit. Neither the uniform identification of varying physical spaces nor the actual bounds (31) follows from the flat gap. Derivatives of $P_f, P_c$, the current, and the nonlinear constraint cannot be absorbed into an unproved constant.

### 2.4 The required scalar–physical cancellation

The separate physical determinant has a growing response even along constant commuting final link angles. The actual globally based finite identity and the local Ward limit give, at fixed $K$,

$$\lim_{N\to\infty} N^{-4}[t^2] \log\det Q_\perp(tA) = C_H|A|^2(K^2 - 1)g_4, \qquad g_4 = \int_{[-\pi,\pi]^4} \frac{dp}{(2\pi)^4\, 4\sum_\mu \sin^2(p_\mu/2)} > 0. \tag{36}$$

Here the fine angles are $tA/K$. This is the separately established flat-holonomy calculation: the same term in the scalar orbit-density ratio cancels (36) in the actual total correction. It is a zero-momentum test, not a magnetic Maxwell coefficient.

Accordingly, the spectral theorem does not permit an independent replacement of a massless physical or scalar factor by its massive infinite reference. A valid curvature comparison must preserve the coupled cancellation, or supply a justified joint infrared regulator. No actual vector logarithmic coefficient, its sign, or Yang–Mills existence conclusion is proved by this section.

# 3 A covariant moving-reference scalar average

We construct tangent operators for the actual ordinary composite average. The nonlinear constraint and its cutoff are unchanged. The construction uses the internal group-valued transport selected in that ordinary map, rather than an unverified identification with an older axial operator carrying the name $Q'$.

## 3.1 Actual data and the primitive relative correction

Fix odd $\ell \geq 3$, $K = \ell^k$, and compatible nested centered blocks. Coordinates are current-link angles with counting scalar and bond inner products. Use

$$U^g(e) = g(e_-)U(e)g(e_+)^{-1}, \qquad (D_U\phi)(e) = \phi(e_-) - \mathrm{Ad}_{U(e)}\,\phi(e_+).$$

The generator acts on all scalar sites. Scalar parameters have angle units at every level; no spacing factor is inserted into this formula.

Let $P_\ell(U; y, x)$ be exactly the internal group-valued transport entering the selected ordinary map $M_\ell$. The actual definition in Bałaban, CMP109 (1987), pp. 252–254, (0.3),(0.5)–(0.12), supplies local analyticity, internal-block support, and

$$P_\ell(U^g; y, x) = g(y)P_\ell(U; y, x)g(x)^{-1}. \tag{37}$$

At identity its derivative is the same arithmetic contour mean that occurs in the ordinary arithmetic differential. These facts also hold when $P_\ell$ is the selected group mean of coordinate-order transporters; it need not be one axial path.

The pinned deposition specifies this ordinary map at Raw I: 831–845. Its relative contour and ordinary increments occur at 46965–46978, and the moving-reference endpoint correction at 47076–47086. Raw I:23913–23920 describes an older axial/nested scalar average. No equality with every such convention is needed: the following operators are defined directly using the actual $P_\ell$.

Freeze $U$, put $W = M_\ell(U)$, and for a small bond increment $x$ define

$$\begin{aligned} r_U(x; y, z) &= \frac{1}{i}\log\Big[P_\ell(e^{ix}U; y, z)P_\ell(U; y, z)^{-1}\Big], \\ \alpha_U(x; y) &= \ell^{-d}\sum_{z\in B_\ell(y)} r_U(x; y, z), \\ a_U(x; y) &= e^{i\alpha_U(x;y)}. \end{aligned} \tag{38}$$

All relative logarithms vanish at $x = 0$, and $a_U(0) = 1$. Let

$$Z_U x = D\alpha_U(0)[x], \qquad C_U x = \frac{1}{i}\,\frac{d}{ds}\Big|_0 M_\ell(e^{isx}U)W^{-1}. \tag{39}$$

The local corrected output with this frozen reference is

$$\overline{F}_U(x; c) = a_U(x; c_-)^{-1}M_\ell(e^{ix}U)(c)a_U(x; c_+).$$

It equals $W$ at zero. Its left-relative differential is exactly

$$\overline{C}_U = C_U - D_W Z_U. \tag{40}$$

This is a moving-reference tangent identity. It is not an identification with the derivative of the fixed-identity corrected map at $U$.

**Lemma 3.1** (Primitive scalar intertwining)**.** *Define*

$$(B_U\phi)(y) = \ell^{-d} \sum_{z\in B_\ell(y)} \mathrm{Ad}_{P_\ell(U;y,z)}\,\phi(z). \tag{41}$$

*With $r_\ell$ restriction to the new centers,*

$$Z_U D_U = r_\ell - B_U, \qquad \overline{C}_U D_U = D_W B_U. \tag{42}$$

*Proof.* For the fine gauge $e^{is\phi}$, the relative contour derivative from (37) is

$$\phi(y) - \mathrm{Ad}_{P_\ell(U;y,z)}\,\phi(z).$$

Its fine left tangent is $D_U\phi$. Averaging this expression proves the first identity. There is no logarithm Jacobian at this step: its argument is identity at the reference point. Ordinary covariance gives $C_U D_U = D_W r_\ell$. Subtract $D_W Z_U D_U$ using (40). □

### 3.2 Joint covariance and composition

For a fixed site gauge $g$, let $O_E(g), O_S(g), O_c(g)$ denote adjoint frames on fine bond source sites, fine scalar sites, and new centers. Since

$$e^{iO_E(g)x}U^g = (e^{ix}U)^g,$$

the relative contour product at $U^g$ is conjugated by $g(y)$. The similarity-equivariant logarithm therefore gives

$$\begin{aligned} Z_{U^g} O_E(g) &= O_c(g) Z_U, \\ \overline{C}_{U^g} O_E(g) &= O_{E_c}(g)\overline{C}_U, \\ B_{U^g} O_S(g) &= O_c(g) B_U. \end{aligned} \tag{43}$$

Both the reference and the varied configuration are transformed. No multiplicative cocycle in the reference is assumed. Adjoint frames are orthogonal for the real invariant trace pairing, and complex orthogonal for its holomorphic bilinear continuation.

At stage $j$, evaluate $C_j, \overline{C}_j, Z_j, B_j, D_j$ at the ordinary reference

$$U_j = M_\ell^j(U).$$

Set

$$\begin{aligned} C^{(t)} &= C_{t-1}\cdots C_0, \\ \overline{C}^{(t)} &= \overline{C}_{t-1}\cdots \overline{C}_0, \\ B^{(t)} &= B_{t-1}\cdots B_0, \end{aligned} \tag{44}$$

with identity products at $t = 0$. The first is exactly the derivative of the actual ordinary composition. Define $Z^{(0)} = 0$ and

$$Z^{(t+1)} = B_t Z^{(t)} + Z_t C^{(t)}. \tag{45}$$

**Proposition 3.2** (Exact composite tangent identities)**.** *At the final level, writing $C, \overline{C}, B, Z, D_c$ for these products and the final generator,*

$$C - \overline{C} = D_c Z, \qquad \overline{C} D_{\mathrm{all}} = D_c B, \qquad Z D_{\mathrm{all}} = r_K - B. \tag{46}$$

*The operators have exact joint gauge covariance with the original fine and final ordinary reference frames.*

*Proof.* The primitive intertwining gives $\overline{C}^{(t)} D_0 = D_t B^{(t)}$. Suppose $C^{(t)} = \overline{C}^{(t)} + D_t Z^{(t)}$. Then

$$C_t C^{(t)} = \overline{C}_t \overline{C}^{(t)} + D_{t+1}\{B_t Z^{(t)} + Z_t C^{(t)}\},$$

which proves the first two identities by induction. For the third, insert $Z_t D_t = r_\ell - B_t$ and $C^{(t)} D_0 = D_t r_{\ell^t}$ in (45); the two intermediate $B_t r_{\ell^t}$ terms cancel. Finally, (43) telescopes at the ordinary reference backgrounds. These are not the fixed-identity corrected iterates. □

### 3.3 Own-block locality and an exact constant row Gram matrix

For $x \in B_K(y)$, let $x = x_0, x_1, \ldots, x_k = y$ be its unique nested sequence of block centers. The scalar product in (44) has only this one ancestry. Consequently

$$\begin{aligned} B(U; y, x) &= K^{-d} \operatorname{Ad}_{P_{\mathrm{hier}}(U;y,x)}, \\ P_{\mathrm{hier}}(U; y, x) &= P_\ell(U_{k-1}; x_k, x_{k-1}) \cdots P_\ell(U_0; x_1, x_0), \end{aligned} \tag{47}$$

and $B(U; y, x) = 0$ outside $B_K(y)$. Each factor is one group element, even when obtained from a selected group mean of path transports.

Its background dependencies lie in the same final block. A stage contour uses internal stage bonds. Each such ordinary bond reads the two descendant blocks at its endpoints, both inside the parent block by centered nested tiling. Iterating downwards proves that no coefficient of this scalar row reads a fine background edge crossing its final block boundary. The same conclusion holds for all of its derivatives.

The rows of $C$ and $\overline{C}$ read only the two final endpoint blocks. The scalar $Z$ row stays in one block: in (45), $Z_t$ reads internal stage bonds, and $C^{(t)}$ on these bonds reads only descendants inside their parent block. This is full coefficient locality, not merely support at flatness.

**Theorem 3.3** (Covariant block projection)**.** *For every real or complex background in the common germ,*

$$B B^T = K^{-d} I. \tag{48}$$

*In dimension four,*

$$P_B = I - K^4 B^T B \tag{49}$$

*is the symmetric projector onto* $\ker B$*, orthogonal for real fields. It is holomorphic, has one-final-block matrix and background support, and obeys*

$$P_B(U^g) = O_S(g) P_B(U) O_S(g)^T. \tag{50}$$

*The normalized term* $\frac{1}{2} \log\det B B^T$ *vanishes identically.*

*Proof.* Distinct rows have disjoint scalar supports because the final blocks partition the fine sites. In the invariant bilinear pairing, $\operatorname{Ad}_P \operatorname{Ad}_P^T = I$. Equation (47) therefore gives

$$(B B^T)_{yy} = K^{-2d} \sum_{x \in B_K(y)} I = K^{-d} I,$$

with zero off-diagonal blocks. This algebra remains valid on the complexified group. The projector assertions follow directly, and covariance follows from that of $B$. □

### 3.4 Flat values and the analytic domain

At identity, the contour derivative in $Z$ is the root term of the actual ordinary arithmetic differential. Subtraction of the two endpoint terms in (40) leaves precisely

$$(\overline{C}_1 x)(c) = \ell^{-d} \sum_{z \in B_\ell(c_-)} x([z, z+\ell e_\mu]), \qquad B_1 = Q_{0,\ell}. \tag{51}$$

Centered-block tiling then gives

$$\overline{C}(1) = A_K^{\mathrm{st}}, \qquad B(1) = Q_{0,K}. \tag{52}$$

The bond map here is an angle map, with a primitive factor $\ell$ relative to the normalized physical-potential map. There is no such factor for the scalar average.

At fixed $K$, each coefficient is a composition of local analytic maps reading finitely many links in one or two final blocks. A small link-angle product ball therefore works uniformly in volume and in unrelated preceding depth $M$. Its radius and constants may depend on $K$. Cauchy derivatives preserve the same support. Compatible tori are read with the prescribed local lifts and nested partition.

The previously proved ordinary-reference trajectory gives a stronger optional bound. For $U = e^{iA/K}$ in its small globally bounded physical-potential ball, stage angles are $O(\ell^j \|A\|_\infty / K)$. Fixed-$\ell$ local analytic Schur bounds and the sum $\sum_{j<k} \ell^j/K \le (\ell-1)^{-1}$ imply

$$\|B - Q_{0,K}\|_{\mathrm{count},\,2\to 2} \le C \|A\|_\infty K^{-2} \qquad (d=4). \tag{53}$$

The corresponding counting scale for the angle bond map $\overline{C}$ is $K^{-1}$. These stronger estimates are not needed for fixed-$K$ volume uniformity or the exact identity (48).

The induced scalar map of the fixed-identity corrected composition is a different operator. It contains logarithm Jacobians, and its row Gram matrix need not be constant. The present moving-reference construction avoids that issue; it must not be replaced by the fixed-identity derivative without the appropriate endpoint conversion.

### 3.5 A basis-free scalar completion

Let $G_{\mathrm{all}} = D_{\mathrm{all}}^T D_{\mathrm{all}}$, and let $V$ be any bilinear orthonormal frame of $\ker B$. Define

$$G_B = V^T G_{\mathrm{all}} V, \qquad \widehat{G}_B = P_B G_{\mathrm{all}} P_B + K^{-2}(I - P_B). \tag{54}$$

The two projector subspaces reduce $\widehat{G}_B$; hence

$$\det \widehat{G}_B = \det G_B \, (K^{-2})^{\operatorname{rank} B}. \tag{55}$$

The extra factor is background independent. The ambient matrix $\widehat{G}_B$ is exactly gauge covariant, without a need to choose gauge-covariant frames $V$.

For the companion vector matrix

$$Q_B = \mathcal{H} + D_{\mathrm{all}} P_B D_{\mathrm{all}}^T + a \overline{C}^T \overline{C},$$

the actual residual covariance and (50) give the same conclusion. When the general tangent determinant identity and the needed spectral gaps are supplied separately, these observations permit a formulation using the two ambient matrices $\widehat{G}_B, Q_B$, with no isolated global base site and no normalized row-Gram term.

Locality and covariance alone do not prove those spectral gaps or the determinant identity. Once they hold, a fixed-$K$ uniform complex perturbation neighborhood permits the usual diagonal logarithm argument. This does not compute a large-$K$ vector Maxwell coefficient or establish a continuum Yang–Mills theory.

# 4 A local block-mean gauge for the actual determinant

This section first proves a finite-dimensional identity, then continues it jointly through the flat field. Its scalar gauge has zero block mean, rather than a point fixed in each block. The resulting scalar and vector operators have flat gaps of order $K^{-2}$ in current counting units. The local analytic continuation is uniform in volume and old depth for each fixed $K$; no uniform large-$K$ curvature comparison is asserted.

## 4.1 Actual residual data and an auxiliary coupled matrix

Fix $K = \ell^k$, old depth $M$, and a compatible finite torus in four dimensions. Let $E, E_c$ be current and final bond-angle spaces. In this section $S, S_c$ contain *all* fine and coarse scalar sites, with no global base removed. Use counting measures and trace-orthonormal color coordinates. Initially work on a real open set where the left-relative gauge generators $D : S \to E$ and $D_c : S_c \to E_c$ are injective. The actual ordinary constraint differential $C$ and center restriction $r : S \to S_c$ satisfy

$$CD = D_c r, \qquad D_b = D|_{\ker r}. \tag{56}$$

Write the full Lagrangian, coarse and residual Hessians as

$$L = s'' - \lambda \cdot f'', \qquad q = s''_{\text{eff}}, \qquad \mathcal{H} = L - C^T q C. \tag{57}$$

The actual minimum comparison supplies

$$\mathcal{H} \geq 0, \quad \mathcal{H}D = 0, \quad CH = I, \quad \mathcal{H}H = 0. \tag{58}$$

On $\ker C$ the nullspace is $\operatorname{ran} D_b$. The actual contour differential satisfies $TD_b = -I$. All of these statements retain the constraint-current curvature $\lambda \cdot f''$.

Let $G = D^T D$, $G_c = D_c^T D_c$, and choose orthonormal bond frames $U, U_c$ perpendicular to the corresponding gauge ranges. Put

$$A = U_c^T C U, \quad \mathcal{H}_\perp = U^T \mathcal{H} U, \quad Q_\perp(a) = \mathcal{H}_\perp + a A^T A, \qquad a > 0. \tag{59}$$

The actual contour/pivot Gaussian amplitude $I_{\text{act}}$ obeys

$$I_{\text{act}}^2 = \frac{\det G}{\det G_c} \frac{a^{n_c - s_c}}{\det Q_\perp(a)}, \qquad n_c = \dim E_c, \quad s_c = \dim S_c. \tag{60}$$

Indeed the constrained Gaussian change of variables $x = Uu + D(\lambda_b, \lambda_c)$ has Jacobian $\sqrt{\det G}$, while the target split has Jacobian $\sqrt{\det G_c}$. The target delta fixes $\lambda_c$, and $TD_b = -I$ integrates $\lambda_b$ with no background determinant. The remaining physical constraint has $AQ_\perp^{-1}A^T = a^{-1}I$, by (58). This proves (60) with the actual target delta in left-relative coordinates; no additional target Haar factor is inserted.

A preliminary projection-free form is

$$\mathbb{B}_a(C) = \begin{pmatrix} \mathcal{H} + DD^T + aC^T C & \sqrt{a}\, C^T D_c \\ \sqrt{a}\, D_c^T C & G_c \end{pmatrix}, \qquad I_{\text{act}}^2 = a^{n_c - s_c} \frac{(\det G)^2}{\det \mathbb{B}_a(C)}. \tag{61}$$

The ratio is first defined on the injective open set. Its normalized logarithm is the joint removable amplitude $\mathcal{D} = \log I_{\text{act}}(W) - \log I_{\text{act}}(1)$; the individual massless logarithms are not evaluated at the flat field. The gapped formula below supplies their joint continuation. We use $\mathcal{N}_0 F = F(W) - F(1)$ only for functions defined there. Its quadratic form is $x^T \mathcal{H} x + |D^T x|^2 + |\sqrt{a} C x + D_c \psi|^2$. Taking the Schur complement in $G_c$ proves (61). Testing $x = 0$ shows that this coupled matrix alone need not have a volume-uniform gap. If $C = \bar{C} + D_c Z$, the shear $\psi' = \psi + \sqrt{a} Z x$ has determinant one, and hence

$$\det \mathbb{B}_a(C) = \det \mathbb{B}_a(\bar{C}). \tag{62}$$

No inverse of $D_c$ is needed for this equality.

### 4.2 The exact block-mean determinant

Assume that $B : S \to S_c$ is onto and that tangent maps satisfy

$$C - \bar{C} = D_c Z, \qquad \bar{C} D = D_c B. \tag{63}$$

Neither $\bar{C}$ nor $B$ is required to be the differential of a nonlinear pushforward with a prescribed Haar density. Let $V$ have orthonormal columns spanning $\ker B$, and define

$$P = VV^T = I - B^T (BB^T)^{-1} B, \quad G_B = V^T D^T D V, \quad Q_B = \mathcal{H} + DPD^T + a\bar{C}^T \bar{C}. \tag{64}$$

Then

$$I_{\mathrm{act}} = a^{n_c/2} \det G_B \, \det(BB^T)^{1/2} \det Q_B^{-1/2}, \tag{65}$$

and therefore

$$\mathcal{D} = \mathcal{N}_0 \left[ \log\det G_B - \tfrac{1}{2} \log\det Q_B + \tfrac{1}{2} \log\det(BB^T) \right]. \tag{66}$$

Here is a direct determinant proof. In orthonormal fine gauge coordinates $DG^{-1/2}$, the inserted map $DV$ has matrix $G^{1/2}V$, and the target gauge row of $\bar{C}$ is $G_c^{1/2} B G^{-1/2}$. Their product is zero. Split fine gauge coordinates into the range of $G^{1/2}V$ and its orthogonal complement. The first block contributes $\det G_B$; elimination of the second block removes the horizontal coarse-gauge row. Since $P_c \bar{C} = P_c C$, the remaining physical block is exactly $Q_\perp(a)$. Thus

$$\det Q_B = a^{s_c} \det G_B \det G_c \det(BG^{-1}B^T) \det Q_\perp(a). \tag{67}$$

Complete $V$ to an orthogonal frame by $W = B^T (BB^T)^{-1/2}$. The lower inverse block in this frame gives

$$\det(BG^{-1}B^T) = \det(BB^T) \frac{\det(V^T G V)}{\det G}. \tag{68}$$

Combining (60), (67), and (68) proves (65). In particular the scalar determinant has power one, not one half.

There is no need to select a frame of $\ker B$ in the local formula. Set

$$\widehat{G} = PD^T DP + K^{-2}(I - P). \tag{69}$$

The two $P$ subspaces reduce this operator and

$$\det \widehat{G} = (K^{-2})^{s_c} \det G_B. \tag{70}$$

Thus $G_B$ may be replaced by $\widehat{G}$ in (66); the extra factor is background-independent.

### 4.3 Actual moving-reference tangent maps

The following construction uses the selected internal contour mean $P_\ell(U; y, x)$ appearing in the actual ordinary primitive average. Its needed properties are local analyticity, endpoint covariance, and arithmetic differential at the identity. No identification with a different axial contour convention is made. Use the gauge sign

$$(D_U \lambda)(e) = \lambda(e_-) - \mathrm{Ad}_{U(e)} \, \lambda(e_+).$$

Let $r_x(U; v)$ be the left differential of $P_\ell(U; y, x)$ in the bond direction $v$, and define

$$Z_U v(y) = \ell^{-d} \sum_{x \in B_\ell(y)} r_x(U; v), \qquad B_U \lambda(y) = \ell^{-d} \sum_{x \in B_\ell(y)} \mathrm{Ad}_{P_\ell(U;y,x)} \, \lambda(x). \tag{71}$$

Endpoint covariance gives $Z_U D_U = r - B_U$. If $C_{\mathrm{ord},U}$ is the actual ordinary angle differential and $C_{\mathrm{mov},U} = C_{\mathrm{ord},U} - D_{M_\ell(U)} Z_U$, then

$$C_{\mathrm{ord},U} = C_{\mathrm{mov},U} + D_{M_\ell(U)} Z_U, \qquad C_{\mathrm{mov},U} D_U = D_{M_\ell(U)} B_U. \tag{72}$$

These formulas follow directly from the actual map and its selected contour mean; they do not modify its nonlinear constraint.

Evaluate these maps at the ordinary intermediate backgrounds $U_j = M_\ell^j(U)$ and put

$$\bar{C} = C_{\mathrm{mov},k-1} \cdots C_{\mathrm{mov},0}, \qquad B = B_{k-1} \cdots B_0. \tag{73}$$

Induction in (72) proves (63); the endpoint remainder obeys $Z_{\mathrm{tot},j+1} = B_j Z_{\mathrm{tot},j} + Z_j C_{\mathrm{ord},j-1} \cdots C_{\mathrm{ord},0}$. Nested blocks give each fine scalar a unique parent chain. Hence

$$B(y,x) = K^{-d} \operatorname{Ad}_{P_{\mathrm{hier}}(U;y,x)} \quad (x \in B_K(y)), \qquad BB^T = K^{-d} I. \tag{74}$$

The row supports are disjoint. Adjoint transports preserve the color inner product, and their bilinear complex continuations preserve the same identity. At flatness,

$$B(0) = Q_{0,K}, \qquad \bar{C}(0) = A_K^{\mathrm{st}}, \qquad P(0) = P_{\ker Q_{0,K}}. \tag{75}$$

The second identity follows by subtracting the two endpoint contour terms from the ordinary flat differential at each primitive step; the remaining straight-string averages compose exactly.

Every map above has fixed-$K$ local coefficients. They transform covariantly under fine and coarse site gauges, because each defining contour transforms at its two endpoints. Consequently $P$ is a covariant scalar projection, $\widehat{G}$ a covariant scalar operator, and $Q_B$ a covariant bond operator. The scalar normalization in (74) is constant. Formula (66) becomes

$$\boxed{\mathcal{D} = \mathcal{N}_0 \left( \log\det \widehat{G} - \tfrac{1}{2} \log\det Q_B \right).} \tag{76}$$

This equality still refers to the original ordinary constraint and its original contour/pivot normalization.

### 4.4 Continuation across stabilizers and flat coercivity

For $SU(N)$ the all-site injective locus is nonempty arbitrarily near flatness. Choose two coarse cycle holonomies $e^{i\epsilon X}$ and $e^{i\epsilon Y}$, with $X$ distinct diagonal and with the nonzero offdiagonal graph of the Hermitian matrix $Y$ connected. For small nonzero $\epsilon$ their common Lie centralizer is zero. Distributing these holonomies over their cycles gives arbitrarily small coarse backgrounds with injective $D_c$. If $D\lambda = 0$, then (56) implies $D_c r\lambda = 0$; thus $r\lambda = 0$, and parallel transport from one center proves $\lambda = 0$. Injectivity is open at each finite regulator. The same argument works for compact semisimple groups by choosing generators with zero common Lie centralizer. An abelian factor requires a separate stabilizer treatment and is not included in this argument.

At flatness block Poincare gives

$$G_B(0) \geq cK^{-2} I, \qquad \widehat{G}(0) \geq \min(c,1) K^{-2} I. \tag{77}$$

Let $R_q$ be the uniform flat coarse Hessian bound and choose $a \geq R_q + 1$. Since $qD_c = 0$ at flatness, $C^T q C = \bar{C}^T q \bar{C}$ there. The old-depth action dominates Maxwell, and therefore $Q_B(0)$ dominates

$$d^* d + DP(0)D^* + (A_K^{\mathrm{st}})^* A_K^{\mathrm{st}}. \tag{78}$$

This form has a gap $cK^{-2}$, uniform in $M, K$ and volume. For a concise proof, use final physical units and decompose $v = t + D_\eta \phi$ with $D_\eta^* t = 0$. Transverse alias Poincare controls $\|t\|^2$ by $\|d_\eta v\|^2 + \|Qv\|^2$. For $b = Q_0 \phi$, the scalar energy-minimizing lift $H_s b$ has

$$P\Delta_\eta H_s b = 0, \qquad \|D_\eta H_s b\|^2 \le C \|d_c b\|^2 .$$

The latter follows by retaining the central alias in $(\sum_r |a_r(p)|^2/P_r(p))^{-1}$; its symbol is at most $P_0/|a_0|^2 \le C|d_c(p)|^2$. The zero mode lifts constantly. Block Poincare applied to $\phi - H_s b$ then gives

$$\|D_\eta \phi\|^2 \le C \|d_c b\|^2 + C \|P\Delta_\eta \phi\|^2 .$$

Finally $d_c b = Qv - Qt$ and $Q$ is a physical $L^2$ contraction. Combining these estimates and restoring counting units proves

$$Q_B(0) \ge c_a K^{-2} I. \tag{79}$$

This proves the mean-zero gauge estimate directly; the projection here is not the earlier source projection onto $\Delta \ker Q_0$.

The operators in (76) are invertible at flatness, although $D^T D$ and $D_c^T D_c$ are singular there. Their determinant branches are fixed by positivity. Equality on the nonempty injective real open set therefore extends holomorphically through flatness. This is a joint continuation of the finite actual amplitude, not a limit of its separate massless orbit factors.

### 4.5 The fixed-ratio local limit and its remaining scope

For fixed $K$, the moving-reference coefficients have finite range in the current background $U$. The identity and residual positivity are evaluated on the actual stationary $W$-germ, or its gauge orbit. After substitution $U = U_*(W)$, the source dependence in $W$ is generally exponential, not finite range. The actual classical support bounds give exponentially local $\mathcal{H}$ and its source derivatives, uniformly in old depth and volume. By (74), $P = I - K^d B^T B$ is itself local. The flat gaps permit a common fixed-$K$ product radius and a positive spatial exponent on which $\widehat{G}, Q_B$ have uniformly accretive real parts and bounded weighted resolvents. Their logarithms and local source derivatives then have exponentially summable kernels. This follows, for example, from a resolvent contour around the common positive spectrum and weighted conjugation; it does not assume a weighted Neumann series converges merely because an unweighted one does.

Along any fixed jointly locally convergent classical structural system, compact-source determinant derivatives now pass jointly to the infinite-volume limit. The finite identity (76) identifies their total kernels with those of the actual leading correction before taking that limit. Vacuum constants are removed by $\mathcal{N}_0$. No uniqueness of the classical structural system is inferred. Covariant diagonal-density and affine-Maxwell consequences can be developed from this gapped pair separately, with their explicit source and limit conventions.

The radius and weighted source constants in this paragraph may depend on $K$. Neither the flat gaps nor the exact identity yet prove a large-$K$ uniform curvature estimate, a bounded reference-comparison finite part, or the sign of the actual running coupling. The residual constraint-current term remains present unless a separate on-shell probe has been established.

## 5 Both flat mean-zero-gauge factors have a physical-scale gap

This section proves spectral estimates for the explicit flat matrices of the corrected all-site gauge construction. The exact nonlinear determinant identity and its continuation through the reducible flat background are separate inputs. No derivative estimate for a varying-background heat trace is inferred from these flat bounds.

### 5.1 Spaces and flat operators

Work in dimension $d = 4$, on a cubic current torus of side $KN$, with $K = \ell^k$ odd, final centers $Ky$, and the disjoint centered cubes $B_y$ of side $K$. All scalar and bond norms are counting norms. Let $D$ be the full-site scalar gradient, $d$ the curl, and $\Delta_s = D^T D$. Let $Q_0$ be the normalized scalar block average, and put

$$P = I - Q_0^T (Q_0 Q_0^T)^{-1} Q_0, \qquad \Pi = I - P. \tag{80}$$

Thus $P$ projects onto block-mean-zero scalars, and $\Pi$ onto block-constant scalars. Choose a block-diagonal orthonormal frame $V$ for $\ker Q_0$ and define

$$G_B = V^T \Delta_s V. \tag{81}$$

At $K = 1$ this scalar space is empty; its trace is zero and its determinant is one.

The actual flat corrected composite differential is the straight angle average $A = A_K$. It obeys

$$AD = D_c Q_0, \qquad \|A\|_{2\to 2} \le K^{-1}. \tag{82}$$

Here $D_c$ is the coarse scalar gradient. The estimate is for this corrected differential; it is not asserted for the full ordinary unprojected constraint.

Let $L = q_M$ and $q = q_{MK}$ be the actual flat classical Hessians. The established homogeneous inputs are

$$L \ge d^T d, \qquad LD = 0, \qquad 0 \le q \le R_q I, \tag{83}$$

uniformly in old depth $M$, endpoint $K$, and volume. Choose a fixed $a \ge R_q + 1$, and set

$$Q_B = L - A^T q A + DPD^T + a A^T A. \tag{84}$$

Its energy satisfies

$$\langle x, Q_B x \rangle \ge \|dx\|^2 + \|PD^T x\|^2 + \|Ax\|^2. \tag{85}$$

Matching these matrices to the nonlinear bridge uses only its flat identifications $B(1) = Q_0$ and $\widehat{C}(1) = A_K$.

### 5.2 Scalar Poincare and a block-mean harmonic lift

If $Q_0 \psi = 0$, the Neumann cube Poincare inequality, summed over the blocks, gives

$$\|\psi\|^2 \le C_P K^2 \|D\psi\|^2, \qquad C_P = 1. \tag{86}$$

Indeed the first nonzero cube eigenvalue is $4 \sin^2(\pi/(2K)) \ge 4/K^2$, and cube-internal edges are a subset of all edges. The case $K = 1$ is vacuous. In particular,

$$G_B \ge C_P^{-1} K^{-2} I. \tag{87}$$

For a coarse scalar $b$, let $H_b$ minimize $\|D\phi\|^2$ subject to $Q_0 \phi = b$. The constraint is onto and fixes the constant nullspace, so the finite minimizer exists uniquely. Its Euler equation is $P \Delta_s H_b = 0$. Moreover,

$$\|DH_b\|^2 \le C_{\text{lift}} K^2 \|D_c b\|^2, \qquad C_{\text{lift}} = \frac{\pi^2}{4} \left(\frac{\pi}{2}\right)^8. \tag{88}$$

To verify the normalization, lift a coarse frequency $p \in [-\pi, \pi]^4$ to the central fine frequency $p/K$. The scalar average has multiplier, up to a unit phase,

$$a_0(p) = \prod_{\mu=1}^{4} \frac{\sin(p_\mu/2)}{K \sin(p_\mu/(2K))}, \qquad |a_0(p)| \ge (2/\pi)^4.$$

Divide the lifted amplitude by $a_0$. The fine difference obeys

$$|e^{ip_\mu/K} - 1| \le \frac{\pi}{2K}|e^{ip_\mu} - 1|.$$

Counting Parseval multiplies the fine squared amplitude by $K^4$, relative to its coarse counterpart, proving (88). The zero mode is constant and has zero energy. All selected frequencies are legitimate on the $KN$-periodic torus. For real data one may take the real part of a complex feasible lift; this preserves the constraint and does not increase energy, including at coarse Nyquist frequencies.

For arbitrary $\phi$, put $b = Q_0\phi$ and $\psi = \phi - H_b$. Then

$$Q_0\psi = 0, \qquad P\Delta_s\psi = P\Delta_s\phi =: g, \qquad \|D\psi\|^2 = \langle\psi, g\rangle.$$

Equation (86) therefore gives $\|D\psi\| \le \sqrt{C_P}K\|g\|$, and hence

$$\|D\phi\|^2 \le 2C_PK^2\|P\Delta_s\phi\|^2 + 2C_{\text{lift}}K^2\|D_cQ_0\phi\|^2. \tag{89}$$

### 5.3 The full vector gap

**Theorem 5.1** (Uniform flat mean-zero-gauge coercivity)**.** *There is a constant $C_{\text{gap}}$, independent of $M, K, N$, such that*

$$\|x\|^2 \le C_{\text{gap}}K^2\{\|dx\|^2 + \|PD^Tx\|^2 + \|Ax\|^2\}, \qquad Q_B \ge C_{\text{gap}}^{-1}K^{-2}I. \tag{90}$$

*Proof.* Use the orthogonal periodic Hodge decomposition $x = x_T + D\phi$, where $D^Tx_T = 0$. Harmonic one-forms are included in $x_T$. The previously proved transverse alias estimate is

$$\|x_T\|^2 \le K^2\{A_0\|P_cAx_T\|^2 + B_0\|dx_T\|^2\}, \quad A_0 = 2c_\omega^{-2}, \quad B_0 = (1 + 2c_\omega^{-2})/4, \quad c_\omega = (2/\pi)^5. \tag{91}$$

It is a Maxwell/averaging inequality, includes zero momentum, and has no old-depth dependence.

By (82),

$$P_cAx_T = P_cAx, \qquad \|D_cQ_0\phi\|^2 \le 2\|Ax\|^2 + 2K^{-2}\|x_T\|^2.$$

Also $P\Delta_s\phi = PD^Tx$ and $dx_T = dx$. Insert these identities in (89), then add $\|x_T\|^2$ and apply (91). A valid constant is

$$C_{\text{gap}} = \max\{(1 + 4C_{\text{lift}})B_0,\ 2C_P,\ (1 + 4C_{\text{lift}})A_0 + 4C_{\text{lift}}\}. \tag{92}$$

This proves the first inequality in (90); (85) proves the second. □

No point-pinned scalar gap or inverse of a globally based scalar matrix occurs in this argument.

### 5.4 A local tent lift controls the full bond Laplacian

**Lemma 5.2** (Block-constant scalar lifting)**.** *There is a block-local map $E : \operatorname{ran}\Pi \to \{$all scalar sites$\}$ such that*

$$\Pi E = I, \qquad \|E\| \le C_0, \qquad \|DE\| \le C_1/K, \qquad C_0 = 2^{d/2}, \quad C_1^2 = 80d\,2^{d-1}. \tag{93}$$

*These bounds include all periodic multiplicities.*

*Proof.* For $K = 2h + 1 \ge 3$, put

$$f(r) = \frac{K}{h+1}\left(1 - \frac{|r|}{h+1}\right), \qquad -h \le r \le h.$$

Its average is one, its mean square is at most two, and its endpoint value and adjacent differences are at most $4/K$. If $e$ equals $c_y$ on $B_y$, define

$$(Ee)(Ky+r) = c_y \prod_{\mu=1}^{d} f(r_\mu).$$

The block mean is $c_y$, and tensorization gives the stated norm bound. For a fixed direction, internal edges contribute at most $16\,2^{d-1}K^{d-2}\sum_y |c_y|^2$ to the squared gradient norm. Interfaces contribute at most $64\,2^{d-1}K^{d-3}\sum_y |c_y|^2$, because $\sum_y |c_{y+e_\mu} - c_y|^2 \le 4\sum_y |c_y|^2$. Divide by $\|e\|^2 = K^d \sum_y |c_y|^2$ and sum directions. The same estimates count the repeated periodic interfaces when $N = 1$ or 2. For $K = 1$, take $E = I$; the conservative constants still apply. □

For a bond field $x$ and $e \in \operatorname{ran}\Pi$, the equality $Ee - e \in \operatorname{ran} P$ gives

$$\langle \Pi D^T x, e\rangle = \langle x, DEe\rangle - \langle PD^T x, Ee - e\rangle.$$

Thus

$$\|\Pi D^T x\| \le \frac{C_1}{K}\|x\| + (C_0+1)\|PD^T x\|. \tag{94}$$

This step uses only a local lifting and duality.

The full bond Laplacian satisfies $\langle x, \Delta_1 x\rangle = \|dx\|^2 + \|D^T x\|^2$. Combining (94), (85), and (90) proves

$$\begin{aligned} &\Delta_1 \le C_\Delta Q_B, \qquad C_\Delta = 1 + 2(C_0+1)^2 + 2C_1^2 C_{\mathrm{gap}}, \\ &Q_B \ge \frac{1}{2C_\Delta}\Delta_1 + \frac{1}{2C_{\mathrm{gap}}K^2} I. \end{aligned} \tag{95}$$

### 5.5 Uniform heat density and inverse moments for both factors

**Corollary 5.3** (Flat local-gauge heat bounds)**.** *For either $O = Q_B$ or $O = G_B$,*

$$N^{-4}\operatorname{Tr} e^{-uK^2 O} \le C\left\{\frac{K^4}{(1+K^2u)^2} + N^{-4}\right\} e^{-cu}, \qquad u > 0, \tag{96}$$

*uniformly in $M, K, N$. The trace for $G_B$ is in its block-local $\ker Q_0$ frame. In the infinite flat structural system the $N^{-4}$ term is omitted. Both factors obey the inverse-moment bounds and the weighted final-time exponential tail of Theorem 2.1, with $O$ in place of its physical $Q$.*

*Proof.* For $Q_B$, apply eigenvalue min–max to (95). The comparison is the full scalar lattice heat trace, with a fixed rescaling of time and multiplicity $4n_{\mathfrak{g}}$, times $e^{-cs/K^2}$.

For $G_B$, (87) gives

$$G_B \ge \tfrac{1}{2}G_B + (2C_P K^2)^{-1} I.$$

The eigenvalues of the compression $V^T \Delta_s V$ dominate the correspondingly indexed full scalar eigenvalues, by the variational principle. Its heat trace is therefore at most the full scalar heat trace. The gap supplies the same exponential factor.

The scalar heat estimate and the substitution $s = K^2 u$ are exactly those used in Theorem 2.1. Final-block translation invariance identifies the normalized total with one block trace. Coarse Bloch integration proves the infinite version directly. Mellin integration gives the stated inverse powers and tails. □

The basis-free scalar completion used by the analytic determinant bridge is

$$\widehat{G}_B = P\Delta_s P + K^{-2}(I - P). \tag{97}$$

At flatness its two subspaces reduce it. Its heat density in final time is exactly

$$N^{-4}\operatorname{Tr} e^{-uK^2\widehat{G}_B} = N^{-4}\operatorname{Tr} e^{-uK^2 G_B} + n_{\mathfrak{g}} e^{-u}.$$

This extra term satisfies (96), after decreasing $c$: for $u \le 1$, $K^4/(1+K^2u)^2 \ge 1/4$, while for $u \ge 1$ it is at least $1/(4u^2)$, and $u^2 e^{-(1-c)u}$ is bounded when $c < 1$. The normalized inverse moment adds exactly $n_{\mathfrak{g}}$, which is absorbed in every case of (26). Thus the actual basis-free pair $\widehat{G}_B, Q_B$ has the same heat, inverse-moment, and final-time tail bounds.

These estimates concern the explicit flat local-gauge factors. They do not establish uniform relative-form derivatives under an affine curvature source, or identify separate massless factors with independent continuum references. The exact nonlinear determinant bridge and the remaining large-$K$ curvature comparison must retain their own hypotheses and proofs.

# 6 Actual local densities in the covariant mean-zero gauge

Use the covariant tangent construction above, with all-site fine gauge generator $D$, covariant scalar block average $B$, corrected bond differential $\bar{C}$, and

$$BB^T = K^{-4}I, \qquad P = I - K^4 B^T B, \qquad \widehat{G} = PD^T DP + K^{-2}(I-P), \qquad Q_B = \mathcal{H} + DPD^T + a\bar{C}^T\bar{C}.$$

The actual finite determinant identity is

$$D_{M,K} = \mathcal{N}_0\left(\log\det\widehat{G} - \tfrac{1}{2}\log\det Q_B\right). \tag{98}$$

Here $\mathcal{H}$ is the full residual Hessian, not a substitution of the bare Wilson Hessian for $S''_M$. The matrix $P$ projects onto scalar fields with zero covariant block mean. It is not the projection onto fields zero at the block centers.

## 6.1 Gauge covariance and localized logarithms

Let a fine site gauge induce orthogonal adjoint matrices $O_S, O_E, O_c$ on fine scalars, fine bonds, and coarse scalars. The covariant tangent construction gives

$$B(U^g) = O_c B(U) O_S^T, \qquad P(U^g) = O_S P(U) O_S^T, \qquad \bar{C}(U^g) = O_{E_c}\bar{C}(U)O_E^T.$$

Together with the exact covariance of $D, \mathcal{H}$ this proves

$$\widehat{G}(U^g) = O_S\widehat{G}(U)O_S^T, \qquad Q_B(U^g) = O_E Q_B(U) O_E^T. \tag{99}$$

These are also complex-orthogonal identities on the holomorphic germs. No choice of a frame for $\ker B$ appears in them.

Fix $K$. The flat lower bounds for both matrices are $cK^{-2}$, uniformly in old depth and volume. Their flat upper bounds and exponentially localized kernels, and their first-two source kernels after actual stationary substitution, are uniform in those same parameters at this fixed $K$. The block-local scalar projection does not enlarge support beyond the corresponding final block; the bond construction has bounded range in final units. The old action and stationary response retain their established exponential source bounds.

Consequently there is a complex output ball, allowed to depend on $K$, on which both logarithms have source-rooted exponential kernels uniformly in volume and old depth. One direct proof is the logarithm series. Choose $s_0$ larger than the flat upper spectral bound. At flatness $\|I - T/s_0\| < 1$, for either matrix $T$. A sufficiently small exponential conjugation and background perturbation preserve that strict bound. Then

$$\log T = (\log s_0)I - \sum_{n\geq 1} \frac{(I - T/s_0)^n}{n}$$

converges in the resulting weighted matrix algebra. First and second source derivatives cost at most polynomial factors in $n$, absorbed by the geometric decay. Every source remains attached to the matrix product chain. Spending a smaller exponential converts the weighted bounds into absolute rooted sums. The fine fiber dimension is finite at this fixed $K$. No endpoint-independent source constant is inferred from this argument.

Group scalar sites and positive fine bonds by their initial final block $x$, and define

$$\rho_x(W) = \mathcal{N}_0 \left\{ \mathrm{Tr}_{x,S} \log \widehat{G}(W) - \tfrac{1}{2} \mathrm{Tr}_{x,E} \log Q_B(W) \right\}. \tag{100}$$

The finite sum is $D_{M,K}$. Each diagonal color trace is individually gauge invariant by (99). The actual stationary branch under an output gauge changes by a fine gauge; its root-gauge convention therefore does not change these traces. Along the chosen jointly locally convergent structural system, these densities have local infinite limits and absolutely summable polynomial source moments, without introducing a globally based scalar inverse.

In particular, each of the two new factors separately has zero thermodynamic quadratic response to a constant commuting connection, at fixed $K$. Apply its local Ward identity to a linear scalar parameter and pass the absolutely summable Hessian row sum. Thus the growing constant-connection terms of the globally based physical quotient have already canceled in the exact change of variables; they are not discarded by a subsequent limit.

### 6.2 Matching the actual Maxwell coefficient

**Proposition 6.1** (Local density and actual affine coefficient)**.** *In every joint infinite structural system at fixed $K$, the density in (100) satisfies*

$$\beta(D_{M,K}) = 2[t^2]\rho_0(w_\mu(n) = tn_\nu H), \qquad \mu \neq \nu, \quad \tau(H^2) = 1. \tag{101}$$

*The polynomial substitution uses the first two localized Taylor kernels. It is not an evaluation of a bounded-field analytic map at a nonzero unbounded connection.*

*Proof.* The linear color jet of each invariant density is zero for the semisimple color algebra. Its quadratic Ward identity kills gradients in either slot. Exponential source moments extend the identity to linear scalar parameters, so its row and column sums vanish for each fixed bond component. Write $k_0(b, c)$ for that density's Hessian on the chosen component and color. Then

$$[t^2]\rho_0(tn_\nu H) = \frac{1}{2} \sum_{b,c} k_0(b, c) b_\nu c_\nu = -\frac{1}{4} \sum_{b,c} k_0(b, c)(b_\nu - c_\nu)^2.$$

All sums are absolute. Translation covariance and reanchoring identify the last expression with $-\frac{1}{4} \sum_r r_\nu^2 \mathcal{K}_{\mu\mu}(0, r)$, where $\mathcal{K}$ is the actual total Hessian kernel. Its Maxwell moment is $-\frac{1}{2} \sum_r r_\nu^2 \mathcal{K}_{\mu\mu}(0, r)$. This proves (101). □

The actual stationary polynomial jets on this probe are the commuting affine fine connection established in Notebook 33. All gauge-covariant matrices may therefore be computed in that representative. The full effective classical Hessian and the covariant averaging operators remain in $Q_B$; the formula does not replace their curvature derivatives by a continuum operator.

### 6.3 A coupled heat representation with the physical flat scale

Let $\widehat{G}(t), Q_B(t)$ denote the actual affine jets just specified, and set

$$h^{\mathrm{mean}}_{M,K}(s) = [t^2]\left\{\mathrm{Tr}_{0,E}\, e^{-sQ_B(t)} - 2\,\mathrm{Tr}_{0,S}\, e^{-s\widehat{G}(t)}\right\}. \tag{102}$$

The fixed-$K$ logarithm/resolvent identities, or ordered Duhamel with the localized first-two vertices, give the exact representation

$$\boxed{\beta(D_{M,K}) = \int_0^\infty h^{\mathrm{mean}}_{M,K}(s)\,\frac{ds}{s}.} \tag{103}$$

There is no additional coarse determinant in this covariant choice: $BB^T = K^{-4}I$, and the added scalar complement in $\widehat{G}$ contributes only a background-independent determinant.

For precision, this is a fixed-$K$ identity on localized Taylor kernels. The spectral gaps, exponential weighted kernels and finite polynomial vertices control the first-two Duhamel coefficients by an integrable small-time and exponentially decaying large-time envelope. Its constants may depend on $K$. This justifies the logarithm integral, polynomial source substitution and local limit; it does not give a uniform curvature estimate as $K$ grows.

The new scalar and vector factors both have flat spectral scale $K^{-2}$. The point-pinned ghost coefficient $4\alpha_g \log K + O(1)$ of Notebook 34 belongs to a different exact determinant splitting and must not be assigned to $\widehat{G}$. The sum in (103) is the same actual coefficient. Its endpoint-uniform magnetic comparison remains to be proved in these new, local, gauge-covariant variables.

## 7 The remaining comparison

Notebook 34 proves the complete based-ghost Maxwell contribution $4\alpha_g \log K + O(1)$ and the full actual ultraviolet bound. In its original local gauge choice, the remaining vector coefficient must be $18\alpha_g \log K + O(1)$ to obtain the specified continuum coefficient $22\alpha_g \log K + O(1)$.

The physical quotient remains an exact finite identity. Its separate massless factors cannot be replaced by independently regulated determinants: the constant-connection calculation gives an explicit noncommutation and the compensation that such a replacement would discard.

The new covariant block-mean representation resolves that infrared issue algebraically. It is an exact identity for the actual finite amplitude, followed by a common local analytic continuation and a fixed-$K$ local limit. Both factors have the physical flat spectral scale, and their diagonal densities have individual gauge identities. No separately singular globally based factor remains in the new heat formula.

The next estimate is the uniform large-$K$ magnetic comparison for this actual pair. The flat heat bounds do not by themselves bound its curvature insertions. The earlier point-pinned ghost coefficient belongs to the old splitting; the new scalar coefficient must be computed with its covariant mean projection retained. The required total is still $22\alpha_g \log K + O(1)$, uniformly in preceding depth and the required structural systems.

A positive actual block drift would activate the finite transfer and tuning results of Notebook 32. Full Yang–Mills existence also requires the measure matching, mixed and large-field histories, sourced continuum limits, nontriviality, Euclidean covariance, positivity and reconstruction in the construction notebook. Those requirements remain part of the same objective.

## References

[1] *Toward a four-dimensional Yang–Mills construction*, Notebooks 32–34 in this workspace.

[2] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited accounts pinned at commit `e392ef45013a581664bab76b90867170b30fe10e`.